Centaurus Therapeutics

A transformative solution for CKM diseases

Ceramide: The Untapped Target in Cardio-kidney/Metabolic (CKM) Diseases

Centaurus Therapeutics
Series B: Clinical Inflection Round

Investment Thesis

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https://news.mayocliniclabs.com/ceramides-miheart/
(Torta et al. 2024)

Ceramides: A Hidden Driver of Chronic Disease

Ceramides, similar to cholesterol, are a family of lipids with signaling and membrane structural functions.

Overproduction of ceramides – often associated with aging, obesity or genetics – is toxic to energetically-demanding organs, including the kidney, heart, and liver.

Excess ceramide signaling leads to organ injury—CKM disease.

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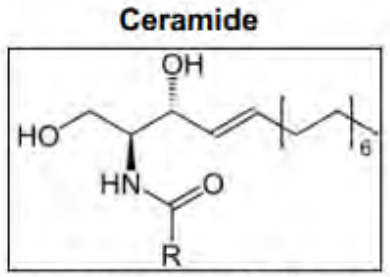
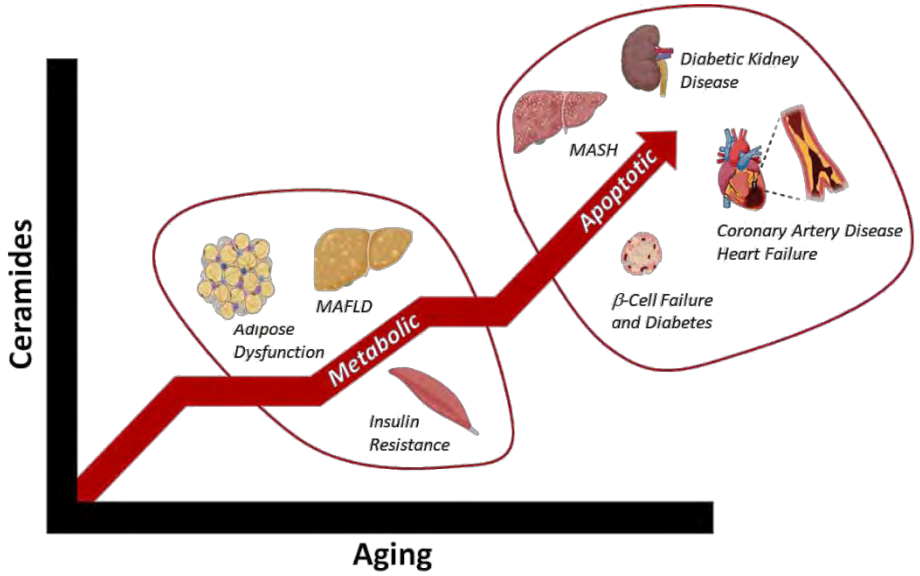


R=alkyl portion of fatty acid



Market Opportunity

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~8–10M

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~1.5–2.0M

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$10B+

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LOREM IPSUM DOLOR SIT:

$30B+

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(Torta et al. 2024)

Competitive Landscape

Pathway Novelty

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Direct Competition

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Centaurus Differentiation

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(Torta et al. 2024)

Phase 1a Healthy Volunteer Trial with CNT2130

Clean Safety Profile

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Excellent Pharmacokinetics

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Robust Pharmacodynamics/DES1 Target Engagement

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Business Overview

Centaurus Therapeutics is a clinical-stage biotechnology company advancing novel therapies for ceramide-driven cardiovascular, kidney, and metabolic diseases, addressing both large and orphan disease opportunities.

Additional therapeutic opportunities available beyond CKM given enormous potential of ceramide biology.

Key Milestones

Lead asset in Phase 1 clinical trial

- Short path to clinical proof-of-concept

Comprehensive preclinical genetic & pharmacology program (animal and ex vivo human tissue studies)

Full IND-enabling safety package

Experienced Leadership and World-Class Scientific Advisors

Core Team

Jeremy T. Blitzer, MD, PhD
Cofounder and CEO/CSO

Hubert Chen, MD
Medical and Regulatory Advisor

Stephen Halasz, JD
Cofounder and Director

Donna Romero, PhD
Medicinal Chemistry

Scientific Co-founder

Scott Summers, PhD
University of Utah

Clinical KOLs & Scientific Advisors

Mark Cooper, MBBS, PhD
Monash University

Samir Parikh, MD
UT Southwestern

David Kelley, MD
Formerly Merck

Jared Rutter, PhD
HHMI / University of Utah

Philipp Scherer, PhD
UT Southwestern

Series B: Clinical Infliction Round

Target Raise:

$XX–XXM

Runway Coverage:

XX-XX months

Primary Objectives:

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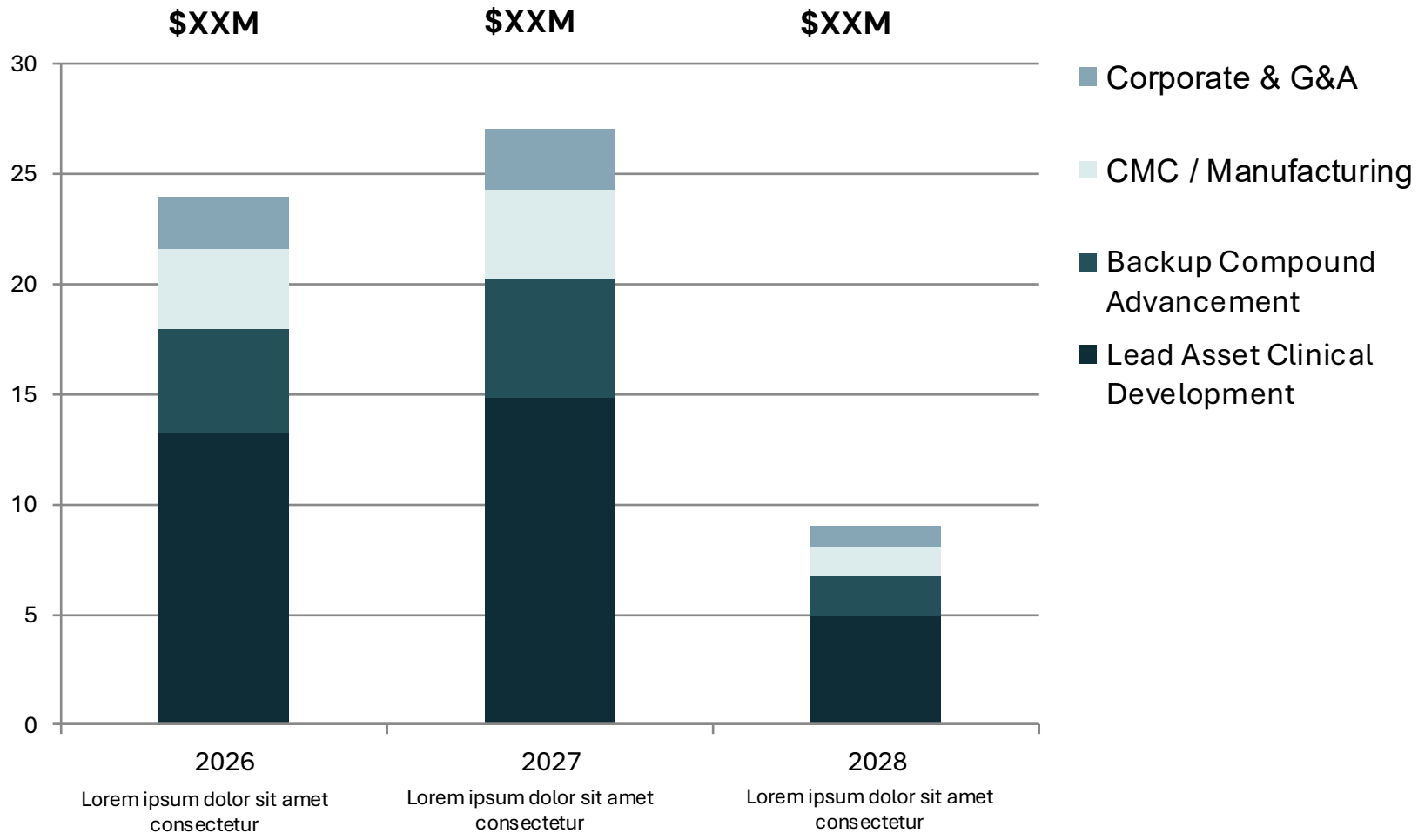
(Torta et al. 2024)

$XXM Financing Advances Lead Asset Through Clinical PoC

Capital allocated to advance lead clinical program through key value inflection points while progressing backup asset and maintaining lean operating infrastructure.



Lead Asset Clinical Development: ~55%

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Backup Compound Advancement: ~20%

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CMC / Manufacturing (Lead Program): ~15%

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Corporate & G&A: ~10%

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Value Inflection Milestones

Clinical

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Pipeline

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Strategic

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(Torta et al. 2024)

Exit & Return Pathways

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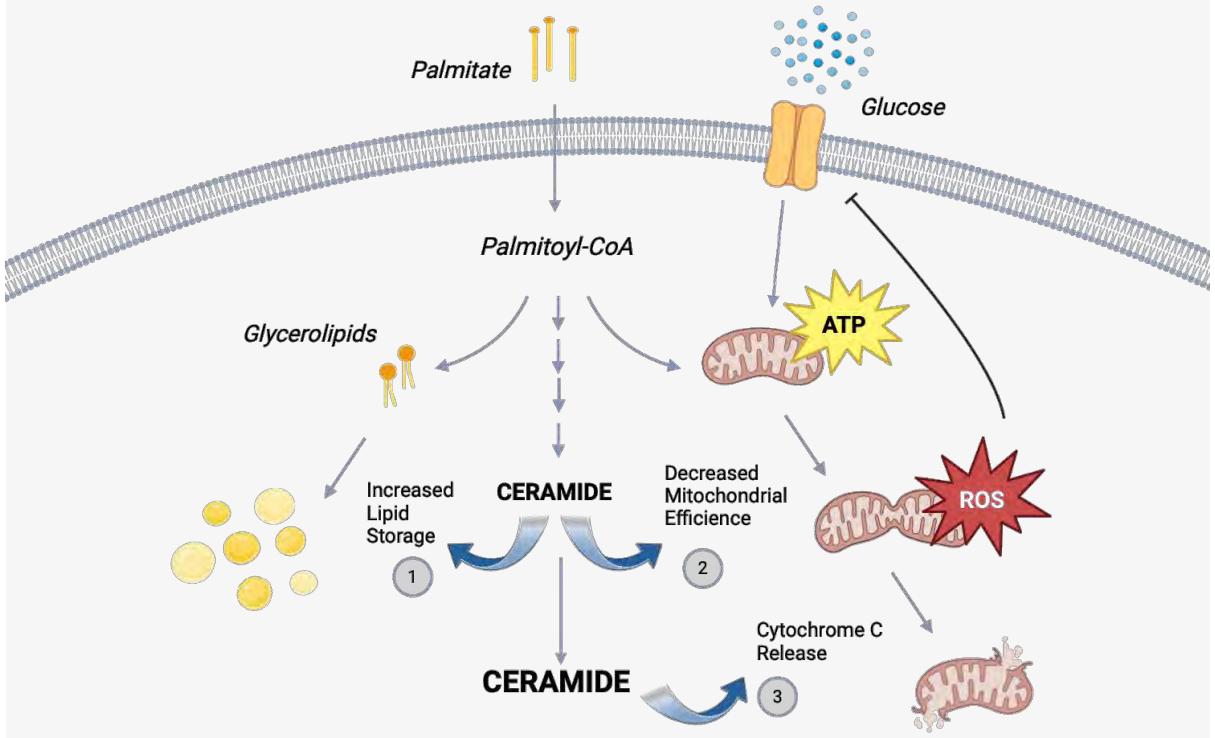
(Torta et al. 2024)

The science

Hyperceramidemia Damages Mitochondria, Driving CKM Diseases

Ceramide overproduction disrupts mitochondria, leading to impaired organ function

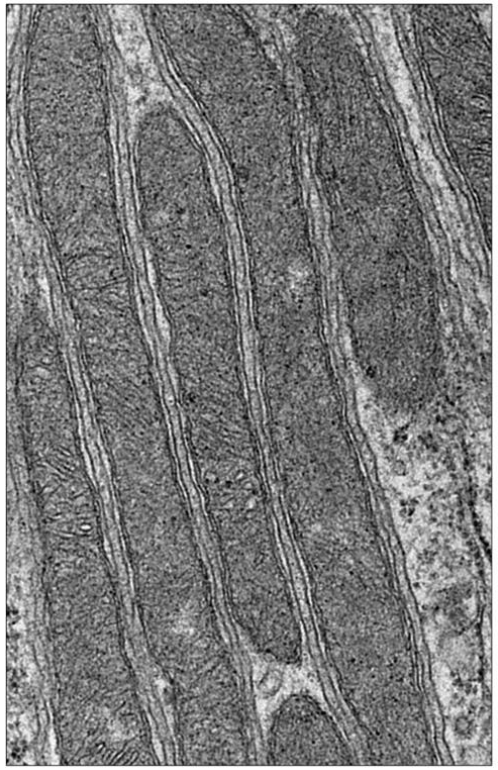
- Ceramides alter cristae dynamics on the inner mitochondrial membrane, reducing respiratory efficiency, and also promote apoptosis (cytochrome C release).
- Injures energetically demanding organs, including kidney, heart, and liver
- Manifests in chronic diseases of the cardio-kidney/metabolic axis



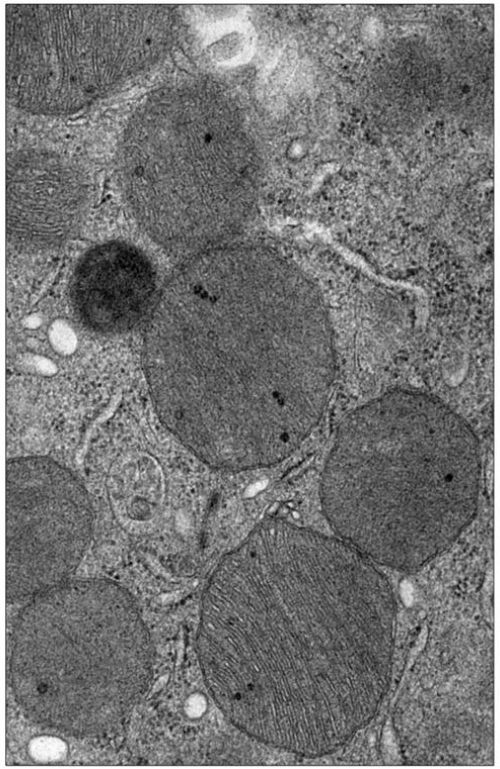
Courtesy of Scott Summers and colleagues

Hyperceramidemia Damages Mitochondria, Driving CKM Diseases

Healthy Mitochondria



Mitochondria in Injured Kidney



Mitochondria in Injured Kidney + CNT Drug

Nickolson et al. (2025) Cell Metab. Epub

Quest Diagnostics Validates Ceramide as a Key Disease Marker

Market Need Confirmed

The availability and clinical adoption of CERT1 underscore growing recognition of ceramides' significance in predicting disease risk and progression.

Therapeutic Opportunity

Centaurus's ceramide inhibitor directly addresses a validated clinical biomarker, aligning with an established medical need.

https://news.mayocliniclabs.com/ceramides-miheart/

(Torta et al. 2024)

Genetic Evidence Confirms Ceramides Drive Disease

Recent genetic studies, including Familial Hyperceramidemia research, establish a direct causal link between elevated ceramide levels and the development and progression of CKM diseases

Gene/Protein	SNP	Ceramide / Kidney Phenotypes
ASAH2 / Neutral Ceramidase	V171L	• Increased ceramides[1] • End-stage kidney disease[1]
ADIPOQ / Adiponectin	G93Q	• Increased ceramides[2] • End-stage kidney disease[2]
CERS2 / Ceramide Synthase 2	E115A	• Increased C16 ceramides in mice[3,4] • Increased HbA1c (hyperglycemia) • Albuminuria / diabetic nephropathy[3]
ACADVL / Very long-chain specific acyl-CoA dehydrogenase	Various	• Increased ceramides[5] • Impaired fatty acid oxidation[5] (cardiomyopathy; rhabdomyolysis; liver dysfunction)

[1]Nicholson, Holland, and Pezzolesi (unpublished) | [2]Simeone et al. (2022) PJ Genom Med 7(1): 43. | [3]Shiffman et al (2014) PLoS One 9, e106631 | [4]Nicholson et al. (2021) JCEM, e3098-e3109 | [5]Norris et al. (submitted)

From Genetic Proof to Therapeutic Action

Genetic Validation

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Pharmacological Translation

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Centaurus drug candidates mimic the genetic DES1 knockout across disease models

Broad Therapeutic Potential Across CKM Disease Indications

Mouse Models	CNT2130 (or analogs)	Inducible DES1 Ablation
DKD (diabetic kidney disease)	✓	✓
HFpEF (heart failure with preserved ejection fraction)	✓	✓
MASH (metabolic dysfunction-associated steatohepatitis)	✓	✓
AKI (acute kidney injury, to simulate injury post-cardiopulmonary bypass)	✓	✓
FAOD (fatty acid oxidation disorder)/Genetically-Driven Cardiomyopathy	✓	✓

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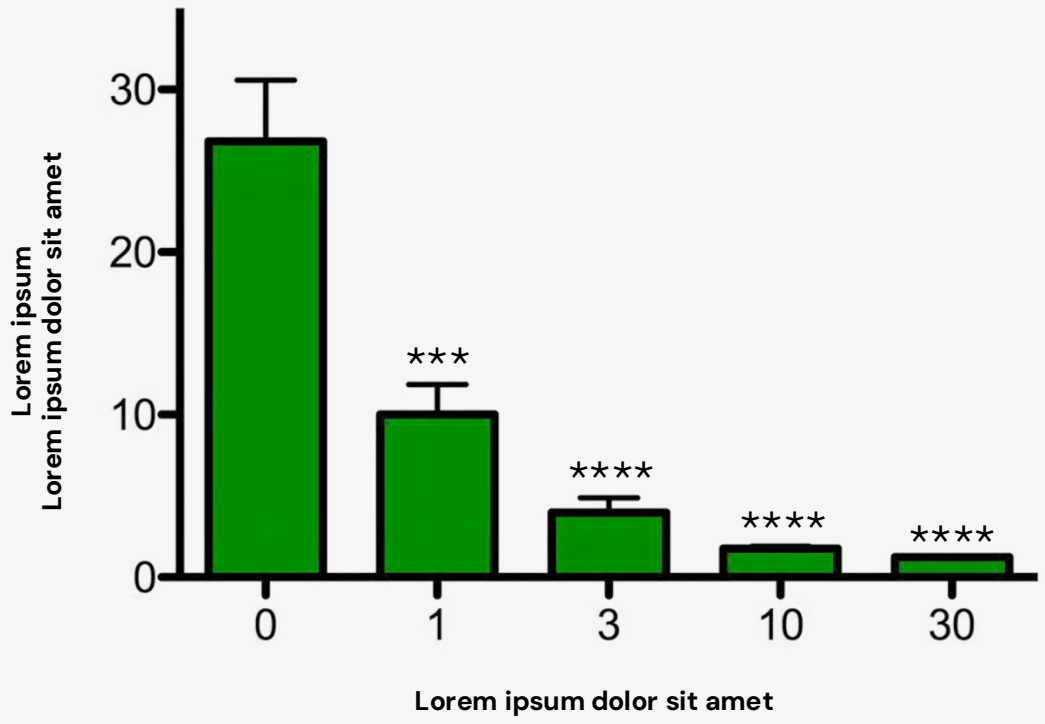

*Allometric scaling assumes 70-kg

CNT2130 Resolves Kidney Injury in Stringent Model of Diabetic Nephropathy

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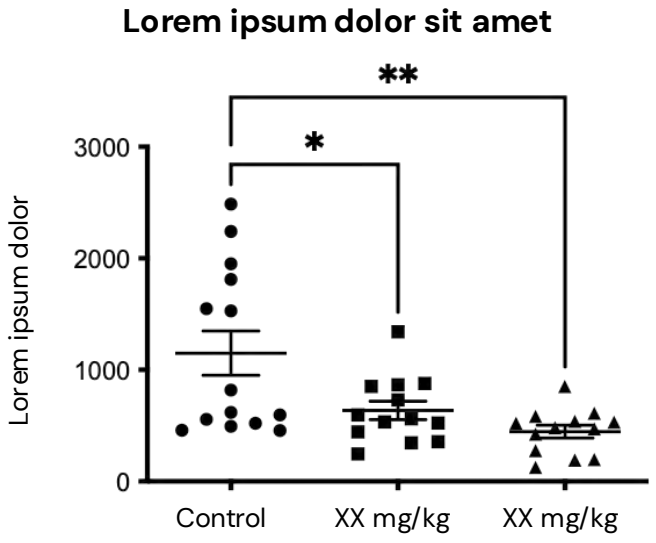

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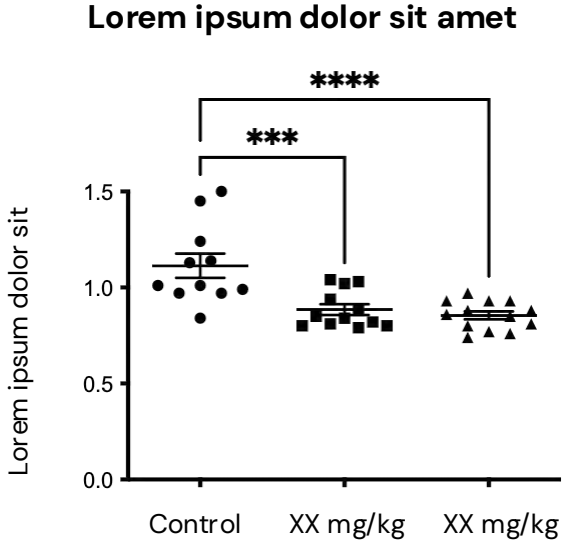

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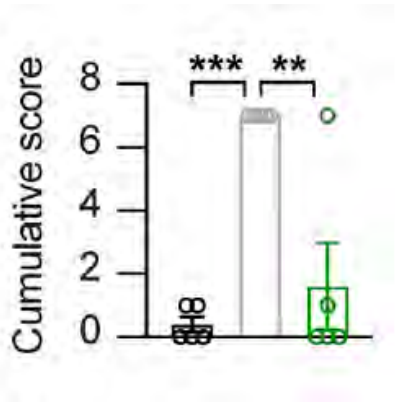

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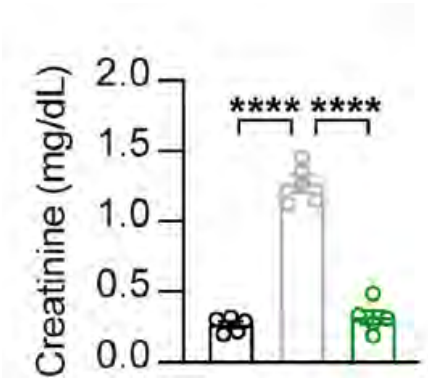

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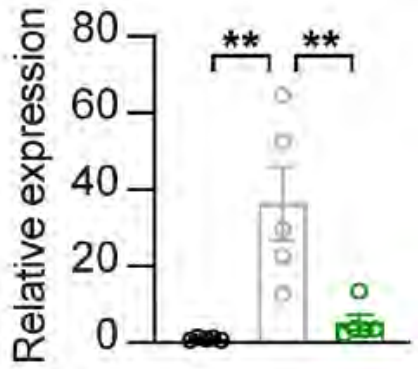

(Nicholson et al. 2025)

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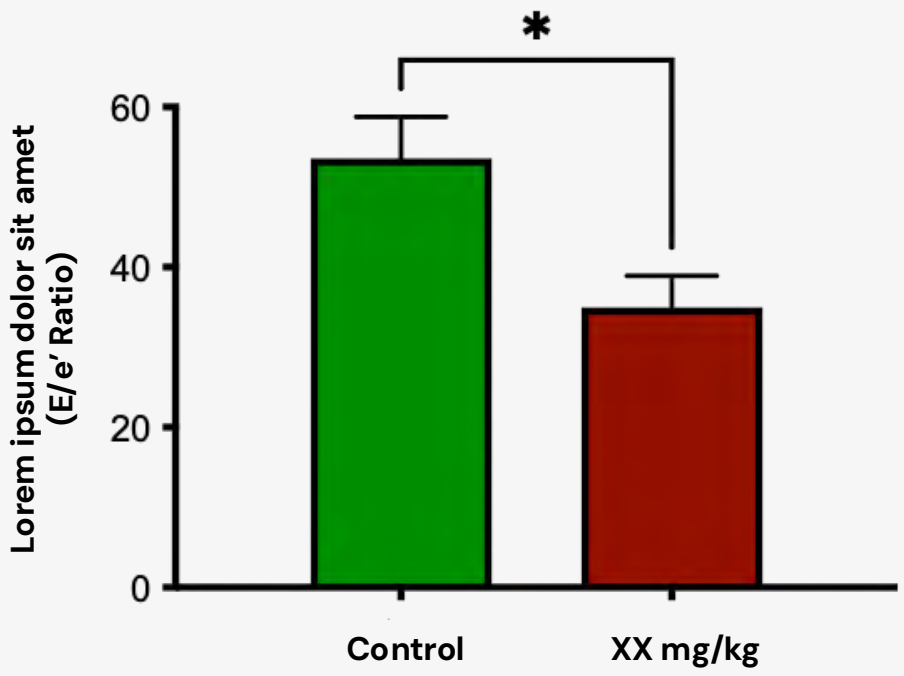

Genetic Validation of Ceramide-Mediated Cardio-Lipotoxicity

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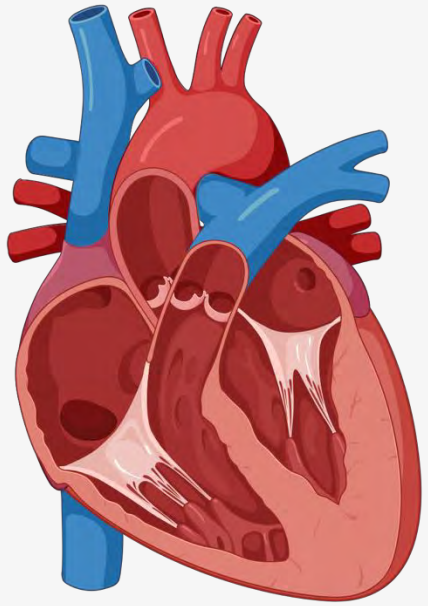


Elevated ceramides

Left ventricular hypertrophy

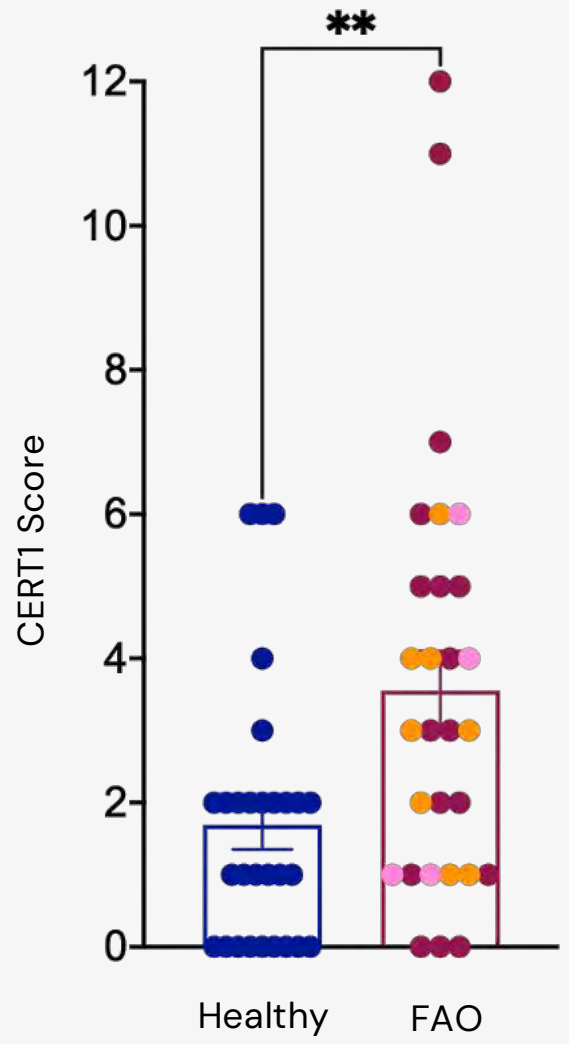
Decreased cardiac hemodynamics

Hyperceramidemia in Patients with Impaired Fatty Acid Oxidation (FAO)

The clinically-validated CERT1 blood ceramide test by Quest Diagnostics confirms significantly higher ceramide levels in patients with fatty acid oxidation disorders.

Analysis includes patients with various mutations affecting FAO

Analysis conducted by Mayo Clinic



(Norris et al. manuscript submitted)

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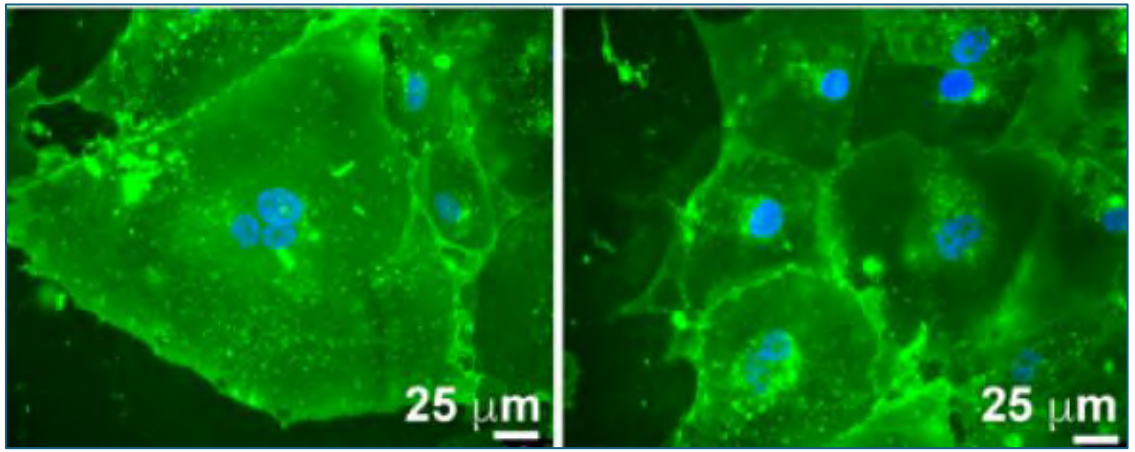

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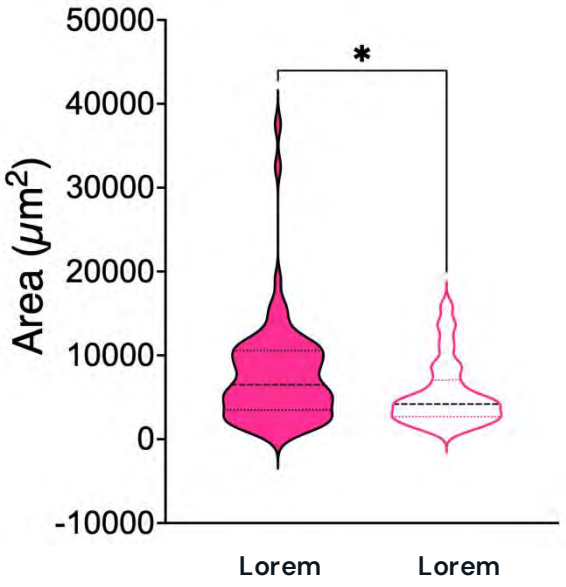

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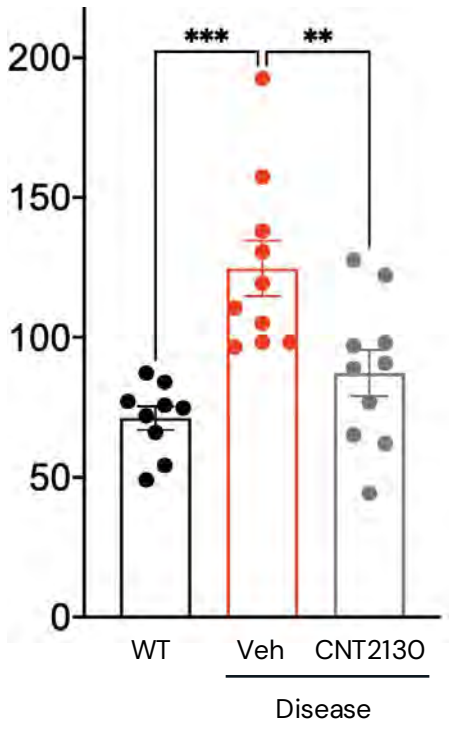

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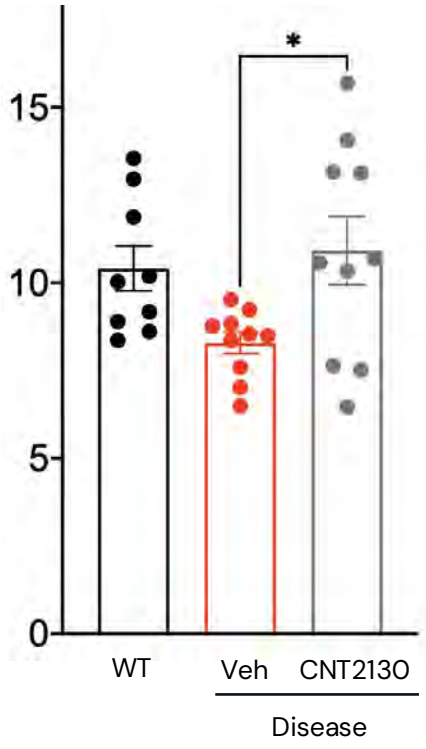

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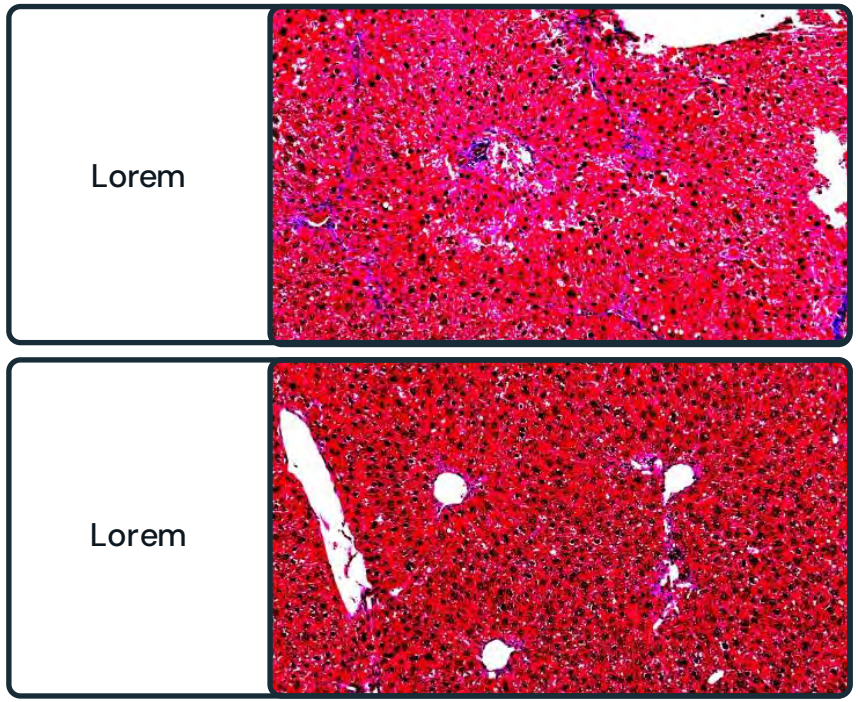

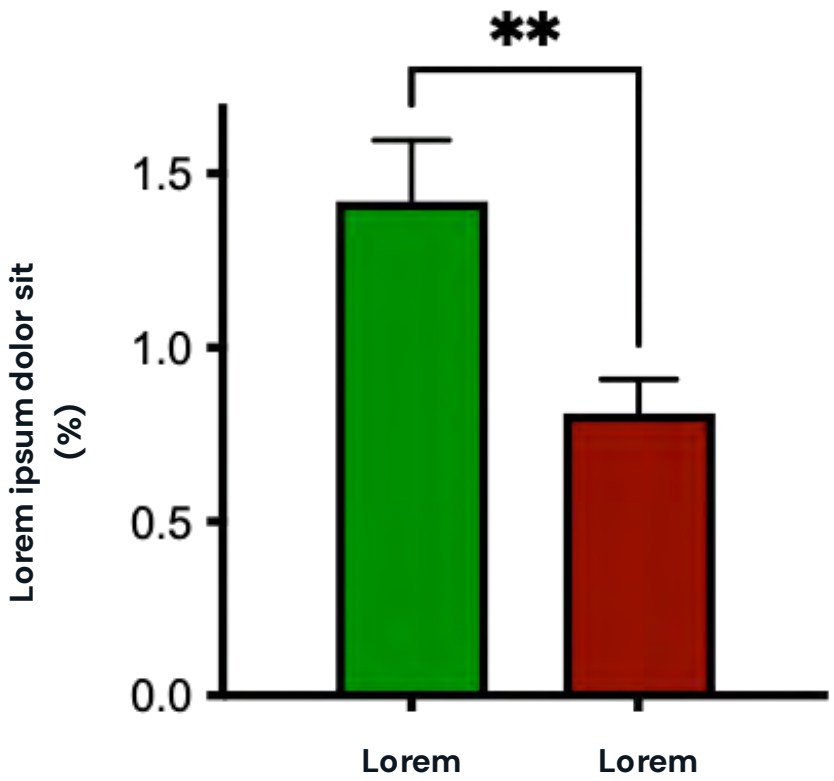

Transformative Candidates with Broad Therapeutic Potential for CKM Diseases

Centaurus's therapeutic candidates—including lead candidate CNT2130—offer clearly demonstrated therapeutic benefits with broad clinical applicability

Kidney

Protection from acute kidney injury

Reduced urine albumin-creatinine ratio

Heart

Improved ventricular diastolic function

Liver

Liver fibrosis reduction/histological improvement

Comorbidities

Reduced blood glucose/triglycerides; pancreatic islet protection

Safety

Well-tolerated with substantial safety margins; target inhibition precedent in humans

Convenience

Oral, once-daily administration

HFpEF: heart failure with reduced ejection fraction
MASH: metabolic dysfunction-associated steatohepatitis

Centaurus Therapeutics: A Transformative Opportunity in CKM Diseases

Centaurus is uniquely positioned to deliver first-in-class, disease-modifying therapies for CKM diseases

Breakthrough Therapeutic Potential

- Enormous potential to revolutionize treatment for CKM syndrome diseases
- Additional orphan disease opportunity in patients with genetically-defined cardiomyopathy (FAOD).

Clinical Stage Asset

- Once-daily disease-modifying oral drug with genetic validation of target/pathway

Robust IP estate Covering Multiple Distinct Compositions that could be Applied to Distinct Indications

Appendix

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A Single Pathway, Multiple Diseases: DES1 Inhibition as a Scalable Therapeutic Platform

Other potential indications based upon Centaurus pharmacology, tool compounds, or genetic knockouts:

Neuro: MS, ALS, AD, stroke

Kidney: FSGS, Alport's disease

Liver: Alcoholic liver disease

Lung: CF, IPF, PPF

Heart: Hypertrophic cardiomyopathy

GI: IBD

Initial Clinical Focus: Targeting Kidney Disease for Rapid Proof-of-Concept

Lead Indication: Diabetic Kidney Disease (DKD)

Rare Disease Opportunity: Orphan Glomerular Diseases

- Focal segmental glomerulosclerosis
- IgA nephropathy
- Lupus nephritis
- Alport syndrome

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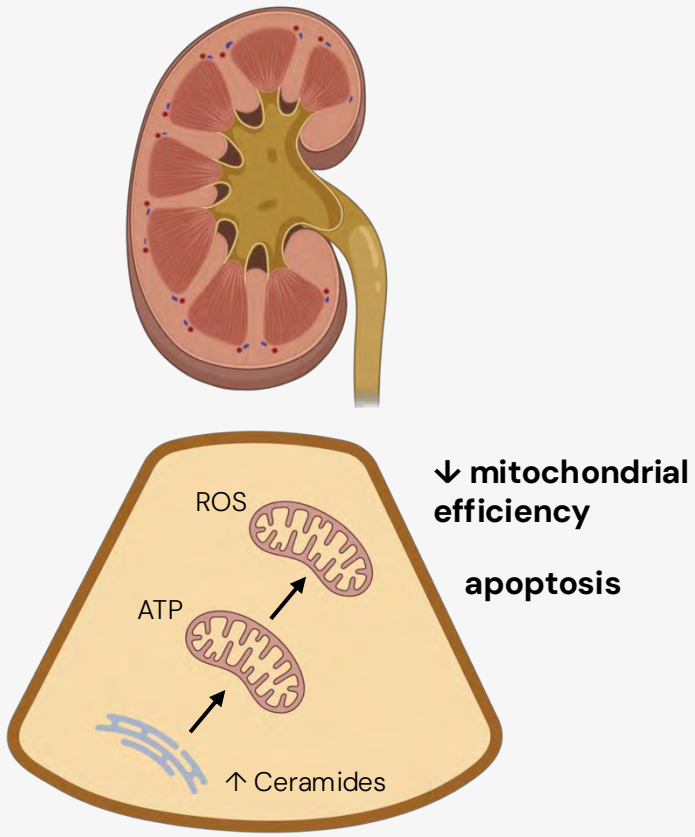
Renal Glomerular/Tubular Injury

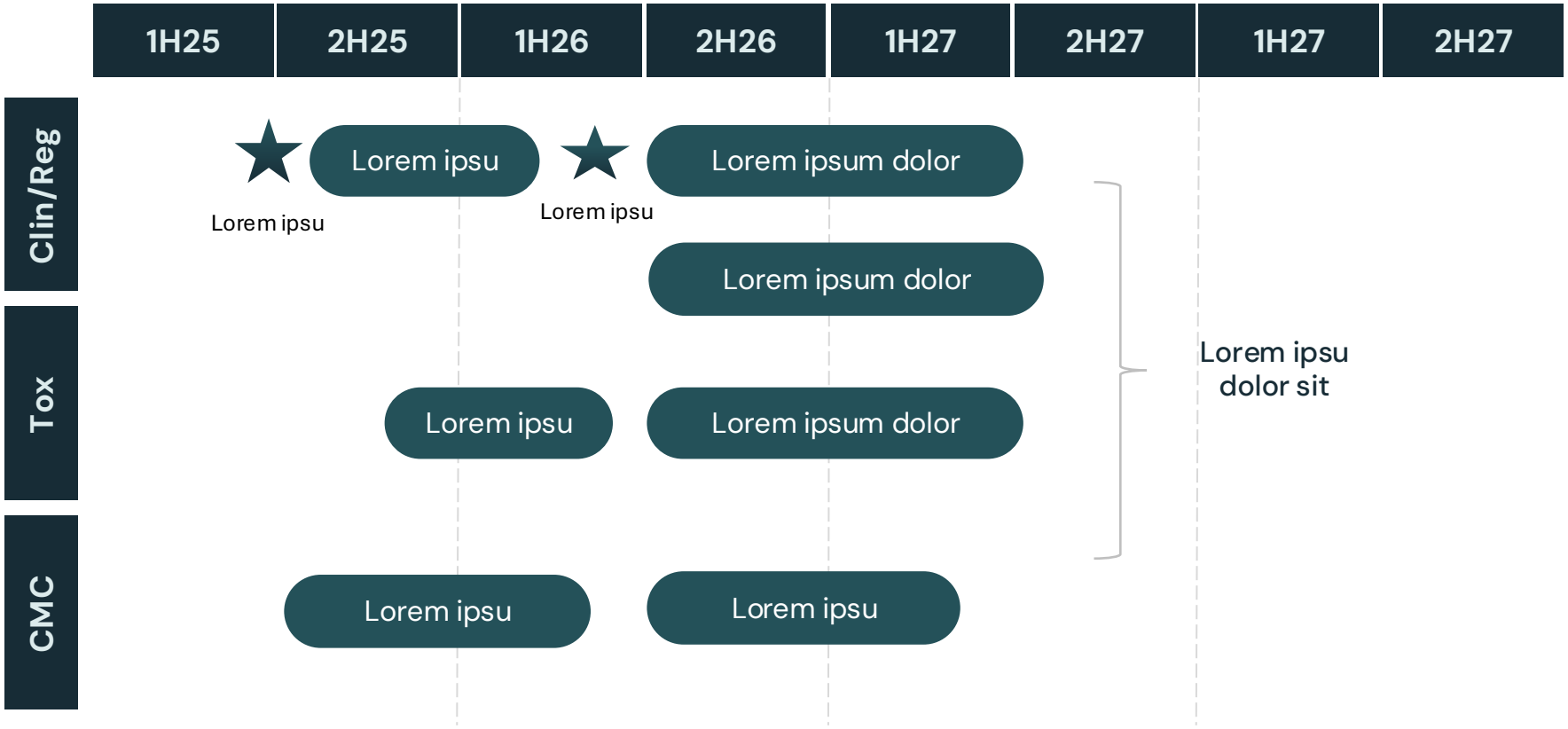


Increased ceramide levels drive diabetic nephropathy and orphan glomerular diseases

*uACR: urine albumin-creatinine ratio

A Well-Defined Path to Clinical Proof-of-Concept

Our clinical plan is designed for rapid validation, with proof-of-concept achievable within months



Strong Safety Profile Validates DES1 Inhibition as a Therapeutic Strategy

Genetic Evidence (Natural Human Validation)

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Long-Term Safety Data

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IND-Enabling Studies

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[1]Blackburn et al. 2019